

Identifiable Marginal Utility: Why Predictor Response Decides When Context Selection Works

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Abstract—Context selection assumes that a candidate’s value can be estimated before the outcome is observed. We study the missing premise: identifiability. We ask what a router can know about the marginal reduction in loss produced by adding a context to the current set, and how that knowledge changes with the fixed predictor it routes into. The analysis gives an exact identity for Bregman losses and two-sided bounds for general smooth losses, with explicit constants for softmax cross-entropy, and shows that conditional on the state the Bayes-optimal squared-loss utility is affine in the predictor’s response. The same identity yields a corollary with practical force: the identifiable part of a candidate’s value is the residual of the fixed predictor itself, so a better predictor leaves less realizable opportunity even when the oracle opportunity is unchanged. We turn this into a measurement protocol — screen-truth correlation, within-episode rank predictability, and a single label-free redundancy statistic — and into a mechanism-based selection rule that adapts to it. Experiments span three regimes. Where utility is estimable (linear subset expert, six traffic benchmarks) learned marginal-utility routing beats fourteen competitors — significantly ahead of the best of them on three benchmarks, unresolved on two and below simple pooling on one — while relevance ranking and maximum marginal relevance are significantly worse than using no context at all on four of the six, and the strongest competitor on every benchmark is the trivial policy of including every candidate. Where it is not estimable (a subset-capable nonlinear expert) no method separates from simply including every candidate, the optimal oracle budget is smaller than the protocol’s, and enlarging the library from sixteen to one hundred and twenty-eight candidates raises the oracle opportunity by forty percent while driving the router to the level of random selection. Where candidates contribute categorical evidence (demonstration selection on five image benchmarks) four different scoring heads bracket the same ceiling. The contribution is a diagnosis and a decision rule, not another ranking heuristic.

Index Terms—marginal utility, identifiability, context selection, negative transfer, predictor response.

I. INTRODUCTION

Prediction systems increasingly have more auxiliary information available than they can afford to use: neighbouring sensors in a sensor network, retrieved documents for a language model, labelled examples for an in-context classifier, candidate feature blocks in a tabular pipeline. Choosing which of them to feed a fixed predictor is usually framed as a ranking problem. Candidates are scored — by relevance, by diversity, by a learned utility model — and the top few are selected.

Ranking presumes that a candidate’s value can be estimated *before* the outcome is observed. This paper studies that premise. We call a candidate’s value *identifiable* at a decision point when it can be recovered, up to useful accuracy, from information the router actually has: the candidate descriptors, the current state, and the frozen predictor’s behaviour. Our

starting point is a decomposition that makes identifiability concrete. For a smooth loss, the marginal value of adding candidate j to the selected set A satisfies

$$m(j \mid A) = -\mathbb{E}[\nabla \ell(f_A, y)^\top d_{A,j}] + O(\mathbb{E}\|d_{A,j}\|^2) - \lambda c_j, \quad (1)$$

where $d_{A,j} = f_{A \cup \{j\}} - f_A$ is the predictor’s *response* to the candidate. The leading term is an inner product between the response and the loss gradient at the current prediction. The response is observable as soon as the predictor is probed; the gradient is not, because it depends on the outcome.

Two consequences organise the paper. First, the response is the right thing to observe: for Bregman losses the decomposition is an exact identity rather than an expansion, for softmax cross-entropy the remainder has a known constant, and for squared loss the conditional expectation is affine in the response — so response features are not a heuristic but the sufficient statistic implied by the loss. Second, the part that must be learned is the predictor’s own *residual*, which means that improving the predictor removes exactly the signal the router needs. A strong predictor is therefore not merely a harder setting for context selection; it is a setting in which the selection problem loses its identifiable content, even though the oracle value is unchanged.

We make this measurable and actionable.

a) Theory.: We give four results. A second-order expansion with a two-sided bound for losses with Lipschitz gradients; an exact identity for Bregman losses that contains the familiar squared-loss expression as a special case; explicit constants for softmax cross-entropy, where the Hessian is positive semidefinite with spectral norm at most $1/2$; and a response-sufficiency statement showing that conditional on the state the Bayes-optimal utility is affine in the response. Together they say that predictor response carries information about marginal utility for any smooth loss, and they say precisely where that information lives.

b) Measurement.: We turn the theory into a protocol that can be run before committing to a selection method: measure how well the cheap cached view correlates with the true marginal, measure within-episode rank predictability from every observable feature, and measure one label-free pool statistic — candidate novelty, the share of a candidate’s variance the anchor does not already explain. Across three regimes these measurements predict, in advance, whether selection will help at all.

c) Decision rule.: The mechanism yields a rule rather than a new heuristic. When candidate novelty is high, retrieval-style ranking is the right policy; when it is low, the best deployable policy is to include everything, because a redundant

pool makes proxy scores actively harmful. We give a label-free adaptive rule with a threshold selected on held-out targets, and we show it converts a harmful default into a beneficial one on two datasets. For settings where the decision can be learned, we describe the response-aware router used throughout, and we show that its head should be shaped by the identity rather than discovered from data.

d) *Evidence.*: The empirical study has three regimes, chosen so that identifiability differs while everything else is held fixed.

- *Utility is estimable.* With a linear subset expert on six traffic benchmarks, learned marginal-utility routing beats fourteen competitors on both datasets examined in detail — significantly ahead of the best competitor on three of six benchmarks, unresolved on two and below simple pooling on one — roughly doubling the strongest non-selection baseline, while relevance ranking, maximum marginal relevance, determinantal diversity, k-center, facility location, k-means representatives and random selection are *significantly worse than using no auxiliary context at all*.
- *Utility is not estimable.* Replacing the expert with a subset-capable nonlinear predictor, with the protocol otherwise unchanged, removes the advantage: no method beats including every candidate, and the oracle’s optimal budget is *smaller* than the protocol’s, so the loss comes from the budget constraint rather than from bad ranking.
- *Candidates contribute categorical evidence.* In demonstration selection for a frozen in-context classifier, four different scoring heads — scalar, curvature-corrected, bilinear and signed bilinear — bracket the same ceiling, which locates the limit in what can be observed rather than in how the score is parameterised.

The remainder of the paper follows those three questions: why response determines utility (Section IV), when utility is identifiable (Section V), and how to select once identifiability is known (Section VI). Section VII reports the three regimes and Section VIII states the boundary and its practical consequences.

e) *Scope and companion manuscripts.*: This paper studies identifiability: what determines whether marginal utility can be estimated at all, and what follows for practice when it cannot. The method itself, its implementation and its engineering evaluation are reported in a companion manuscript, which also carries the multi-benchmark and second-task-family evidence; the present paper reuses that protocol only as an instrument and does not restate it. Neither paper depends on the other for its claims: the identifiability results here are stated and proved from the utility definition, and the experiments are self-contained.

II. RELATED WORK

Auxiliary contexts and demonstrations adapt a frozen predictor to a query [1, 2]. Their quality and ordering matter, and many methods retrieve or select examples before prediction [3–6]. Other work studies prompt-order sensitivity and calibration [7, 8]. Retrieval-augmented systems condition predictions

on external passages and records [9–11], and retrieval scale and adaptation are studied in [12–15]. Recent work combines retrieval with adaptive chains and causal reasoning [16, 17]. Diversity-aware selection penalizes repeated content or maximizes coverage [18–20], and listwise rerankers compare candidate sets directly [21–25]. Long-context studies show that position and context composition change how a model uses retrieved information [26]. These approaches determine which information is relevant or diverse; they do not directly estimate the change in end-task loss produced by adding a candidate to the context already selected.

Outcome-aware selection moves closer to the final prediction objective by valuing examples through their effect on model output. Hashimoto et al. estimate the incremental utility of a demonstration relative to an empty prompt [27]. Data valuation and influence estimation assign credit to training examples through their effect on predictions [28–31]. These methods characterize a candidate’s standalone contribution. Our object conditions on a nonempty, evolving selected set, so the value of the same candidate can change after every routing decision.

Set-selection methods provide the structural counterpart to this decision problem. Submodular objectives formalize coverage, diversity, and diminishing returns [32–34], while determinantal point processes model diverse subsets probabilistically [35]. Adaptive submodularity and influence maximization extend these tools to sequential decisions [36, 37], and streaming or contextual variants address budgeted settings [38, 39]. Classical monotone formulations reward additional elements; context utility in our setting can also be negative because a candidate can increase prediction loss. Negative transfer has been characterized in general transfer learning [40–42] and mitigated in sequential recommendation through multi-modal re-learning [43].

Routing and decision-focused learning treat selection as a learned decision problem. Mixture-of-experts systems route inputs to expert subsets [44–46], and sparse or soft routing architectures reduce computation while increasing capacity [47–50]. Routing networks select computation paths for multi-task learning [51, 52], and recent systems apply dynamic routing across language, vision-language-action, and collaborative-perception tasks [53–55]. Decision-focused learning connects prediction quality to downstream decision quality [56–59], and task-based end-to-end learning follows the same principle [60]. Sequential selection is related to bandit exploration and off-policy value estimation [61, 62]. R-MUR applies this perspective to auxiliary contexts for a fixed predictor, where the routing target is the marginal change in end-task loss and the predictor’s response becomes part of the available decision information.

A. Identifiability and observability of selection decisions

A separate line of work asks not which candidate to pick but whether the decision is determined by the information available at decision time. Sensor placement and submodular selection study when a subset is sufficient for a target and provide the standard criteria — submodular coverage, facility

location and adaptive submodularity [33, 34, 36, 37] — together with diversity criteria such as determinantal point processes and maximum marginal relevance [18, 35]. Section V measures directly whether such criteria track realised utility in a given problem, and regime A shows that they can be not merely uninformative but harmful. In retrieval-augmented generation and dense retrieval the retrieved context is usually assumed to help [9–13, 21]; our pool-geometry experiment identifies the condition under which that assumption inverts. For in-context learning, comparative studies find that similarity retrieval is hard to beat [3, 4], with diversity-oriented variants aimed at coverage [19, 20, 25]; regime C reproduces that finding and explains it through the information a demonstration contributes. Work on data valuation and subset selection for training [28, 29, 31] shares the goal of estimating a candidate’s contribution before committing to it. Closest in spirit is the analysis of routing signals in multi-context prediction [54, 55]; we contribute an exact response-utility identity, a measurement protocol derived from it, and a selection rule that changes with the measured regime.

III. STATE-CONDITIONED CONTEXT SELECTION

Let S_0 be an anchor context, x a query, y its target, and $\mathcal{C} = \{C_1, \dots, C_K\}$ a candidate pool. A selected set is $A \subseteq \{1, \dots, K\}$. A predictor produces $f_A(x)$ from S_0 and the contexts indexed by A ; the predictor is trained before routing and kept fixed during context selection and router training.

We define the net set utility as

$$F(A) = \mathbb{E}[\ell(f_\emptyset(x), y) - \ell(f_A(x), y)] - \lambda c(A), \quad (2)$$

where $c(A)$ is a context cost. The marginal utility of a candidate is

$$m(j | A) = F(A \cup \{j\}) - F(A). \quad (3)$$

For one observed episode, the training target is the counterfactual loss difference

$$\tilde{m}(j | A) = \ell(f_A(x), y) - \ell(f_{A \cup \{j\}}(x), y) - \lambda c_j. \quad (4)$$

This is a realized observation of the marginal. Its interpretation as an expectation requires a fixed predictor and a specified state-sampling rule.

For squared loss, let $\mu = \mathbb{E}[y | S_0, x, \mathcal{C}]$ and let $d_{A,j} = f_{A \cup \{j\}}(x) - f_A(x)$. The marginal separates into

$$m(j | A) = 2d_{A,j}(\mu - f_A(x)) - d_{A,j}^2 - \lambda c_j. \quad (5)$$

The first term is the interaction between the current residual and the direction of the candidate-induced correction, the second term penalizes overcorrection, and the third accounts for cost. Relevance and standalone candidate scores do not determine these three terms, and the same candidate can be useful at one state and harmful at another.

Two oracle policies expose the structural value of this state dependence. The standalone oracle ranks candidates by the true standalone marginal $m(j | \emptyset)$ and keeps that order fixed. The sequential oracle recomputes the true marginal after every addition. Let G_{stand} and G_{seq} denote their prediction

gains over the anchor-only predictor. We measure structural headroom as

$$H_{\text{state}} = \frac{G_{\text{seq}} - G_{\text{stand}}}{G_{\text{seq}}}. \quad (6)$$

This quantity is an evaluation diagnostic. It isolates the value created by conditioning on the selected set, independently of router estimation error.

IV. WHY RESPONSE DETERMINES UTILITY

Throughout, $f_A(x)$ is the prediction of a predictor that consumes the anchor context and the selected set A , y is the outcome, ℓ is the task loss, and $d_{A,j}(x) = f_{A \cup \{j\}}(x) - f_A(x)$ is the predictor’s response to candidate j at state A . The marginal utility of that candidate is

$$m(j | A) = \mathbb{E}[\ell(f_A, y) - \ell(f_{A \cup \{j\}}, y)] - \lambda c_j. \quad (7)$$

The predictor is frozen while utilities are generated, so every statement below is conditional on the state and on the predictor.

A. A two-sided bound for smooth losses

Proposition 1 (Response expansion). *Assume that for every episode $f \mapsto \ell(f, y)$ is twice continuously differentiable on the segment between $f_A(x)$ and $f_{A \cup \{j\}}(x)$, and that $\|\nabla^2 \ell(z, y)\|_2 \leq L$ on that segment. Then*

$$m(j | A) = -\mathbb{E}[\nabla \ell(f_A, y)^\top d_{A,j}] - \frac{1}{2} \mathbb{E}[d_{A,j}^\top \nabla^2 \ell(z, y) d_{A,j}] \quad (8)$$

$$|m(j | A) + \mathbb{E}[\nabla \ell(f_A, y)^\top d_{A,j}] + \lambda c_j| \leq \frac{L}{2} \mathbb{E}\|d_{A,j}\|^2. \quad (9)$$

The leading term is the alignment between the loss gradient at the current prediction and the direction in which the candidate moves the predictor. The remainder is second order in the response magnitude: the first-order term is accurate exactly when the candidate perturbs the predictor little, and a large response can cost at most $(L/2)\mathbb{E}\|d_{A,j}\|^2$. The response is observable at decision time; only the gradient is not.

B. An exact identity for Bregman losses

Proposition 2 (Exact identity). *Let $\ell(f, y) = D_\phi(y, f) = \phi(y) - \phi(f) - \nabla \phi(f)^\top (y - f)$ for convex differentiable ϕ — squared loss, deviance losses, and exponential-family negative log-likelihoods in their natural parametrisation. Then*

$$m(j | A) = \mathbb{E}\left[(\nabla \phi(f_{A \cup \{j\}}) - \nabla \phi(f_A))^\top (y - f_A)\right] - \mathbb{E}D_\phi(f_A, f_{A \cup \{j\}}). \quad (10)$$

Equation (10) is exact and requires no smoothness constant. The second term is a non-positive movement penalty in the Bregman geometry; the first is the inner product between the change in the loss gradient and the current residual. With $\phi(z) = \|z\|^2$ it becomes $\mathbb{E}[2d_{A,j}(y - f_A) - \|d_{A,j}\|^2] - \lambda c_j$, the squared-loss expression used by response-aware routers, which is therefore a special case rather than the whole story.

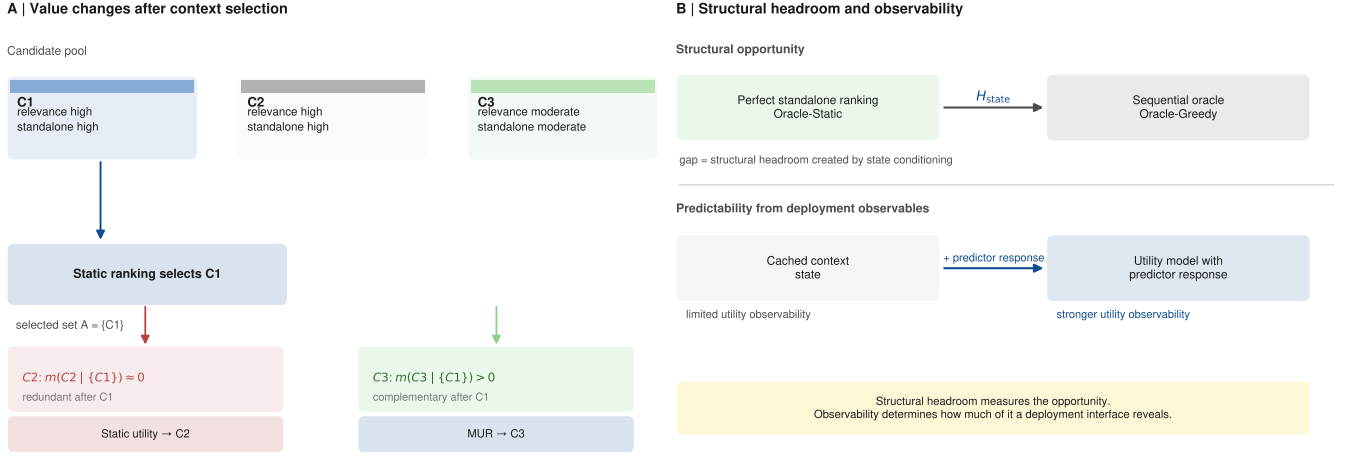


Fig. 1. State-conditioned context selection. A candidate’s standalone value can change after similar information is selected. The gap between a standalone oracle and a sequential oracle measures structural headroom, while the information available at inference time determines how much of that value a router can estimate.

C. Explicit constants for cross-entropy

Proposition 3 (Softmax cross-entropy). *For logits $f \in \mathbb{R}^C$, one-hot y , and softmax cross-entropy $\ell(f, y) = -f_y + \log \sum_k e^{f_k}$ with $p = \text{softmax}(f)$, the Hessian $\nabla^2 \ell = \text{diag}(p) - pp^\top$ is positive semidefinite with spectral norm at most $1/2$. Consequently*

$$\langle e_y - p_A, d_{A,j} \rangle - \frac{1}{4} \mathbb{E} \|d_{A,j}\|^2 - \lambda c_j \leq m(j | A) \leq \langle e_y - p_A, d_{A,j} \rangle + \frac{1}{4} \mathbb{E} \|d_{A,j}\|^2 - \lambda c_j. \quad (11)$$

The upper bound holds because the second-order term penalises any positive semidefinite curvature; the lower bound says that a large response can lose at most a quarter of its squared size in logit space. Both constants are explicit, which matters when the response is used as a feature.

D. Response sufficiency

Proposition 4 (Response sufficiency). *For squared loss, conditioning on the state gives*

$$\mathbb{E}[\tilde{m}(j | A) | x, A] = \langle g_A(x), d_{A,j}(x) \rangle - \|d_{A,j}(x)\|^2 - \lambda c_j, \quad (12)$$

where $\mu(x) = \mathbb{E}[y | x, A]$.

Conditional on the state, the Bayes-optimal squared-loss utility is affine in the response vector plus a known quadratic penalty. State features supply the coefficient $g_A(x)$, response features supply $d_{A,j}$, and the penalty is a function of the response alone. Any score that is independent of the response can only recover the state average and provably omits the interaction $\langle g_A, d_{A,j} \rangle$ that state-conditioned selection exists to exploit.

E. The identifiability corollary

Corollary 5 (What is left to learn). *Under Proposition 4, the only unknown in the optimal score is $g_A(x) = 2(\mu(x) - f_A(x))$, the residual of the fixed predictor. In particular, if the predictor is calibrated so that $\mathbb{E}[y - f_A(x) | x, A] = 0$ for every observable state, the conditional expected utility of*

every candidate collapses to $-\|d_{A,j}(x)\|^2 - \lambda c_j$, which is non-positive and decreasing in the response magnitude.

The corollary is the paper’s central practical statement. A better predictor is one whose residual is closer to unpredictable, so the term the router must estimate shrinks towards zero while the curvature penalty remains. The oracle opportunity — what a router could achieve if it knew the outcomes — does not shrink; what shrinks is the part that is identifiable before the outcome is observed. This is why strong predictors are not simply a harder version of the same problem.

F. From estimation error to routing regret

Theorem 6 (One-step routing regret). *At a fixed state A , suppose the utility estimator satisfies $|\hat{m}(j | A) - m(j | A)| \leq \epsilon$ for every remaining candidate. If $\hat{j}$ maximises the estimate and j^* maximises the true utility, then $m(j^* | A) - m(\hat{j} | A) \leq 2\epsilon$.*

Combining Theorem 6 with Proposition 1 bounds the regret of a response-based router by twice the estimation error of its first-order surrogate plus the second-order response term. The bound is per decision; sequential composition additionally requires that locally accurate choices do not visit a qualitatively different state, which is why Section V measures identifiability directly instead of assuming it.

V. WHEN IS UTILITY IDENTIFIABLE?

Corollary 5 says that the identifiable part of a candidate’s value is the predictor’s residual. That statement is useful only if it can be measured before a selection method is chosen. This section gives three measurements, in increasing order of cost, and states what each one rules out. They are used throughout Section VII.

A. Measurement 1: does the cheap view know anything?

The first question is whether the representations a router would use to screen candidates carry any signal about the true

marginal at all. Given a batch of episodes, compute the true marginals of the frozen predictor at a fixed state and correlate them with the score a cached or response-free model would assign. Two numbers matter: the correlation itself, and the recall of the shortlist a screening stage would form. If the screen’s correlation is indistinguishable from zero, then the recall of a top- q shortlist is at chance level q/K , and no reranking stage downstream can recover the value it never received.

B. Measurement 2: is the ordering recoverable within an episode?

Selection needs an ordering within an episode, not a level across episodes. We therefore fit the strongest available model — gradient boosting or a small network — to the true marginal using every quantity a router can observe: anchor features, candidate features, candidate identity, the response vector, its magnitude, and the observable base–response interaction. We report the out-of-sample R^2 , the mean within-episode rank correlation between predicted and true marginals, an in-sample fit as an alignment control, and a label-permutation control. A negative out-of-sample R^2 together with a within-episode rank correlation near zero is evidence of an identifiability limit rather than of a weak model, because the control establishes that the pipeline itself is sound.

C. Measurement 3: one label-free pool statistic

The third measurement explains when a *simple* policy will work, and it costs nothing: candidate novelty, the share of a candidate’s variance that the anchor does not already explain. Regress each candidate series on the anchor series using the training split and keep the residual variance share,

$$\nu_j = \frac{\text{Var}(c_j - \text{proj}_{\text{anchor}}(c_j))}{\text{Var}(c_j)}. \quad (13)$$

A pool whose candidates have low novelty is redundant with the anchor: retrieval-style ranking then selects candidates that add nothing and can be actively harmful, whereas a pool with high novelty rewards retrieval. The statistic is label-free, needs only the training split, and transfers across datasets because it is a variance share rather than a correlation level.

D. How the three measurements fit together

The measurements are ordered by cost and by what they can rule out. Measurement 1 is a screening sanity check: if it fails, the cached stage is noise and any method that depends on it must be redesigned. Measurement 2 is the identifiability test proper: if it fails, the ordering is not recoverable from observable features and no scoring head will fix it. Measurement 3 is the policy selector: it distinguishes the regime in which retrieval is the right default from the regime in which including everything is. Section VII applies all three and Section VI turns the third into a deployable rule.

Structural headroom, the quantity the protocol measures as the oracle’s advantage over a fixed set, is a necessary condition and not a usable predictor. Within the six-dataset

ridge protocol the per-target correlation between headroom and the router’s advantage over pooling is $+.040$ over 192 targets ($p = .58$), and across datasets the ordering is broken by PEMS03, which has the highest mean headroom (.119) and an advantage of $+.0013$, against PEMS-BAY’s similar headroom (.115) and $+.0303$. The measurement that does discriminate is the *realised* share, computed under the deployment’s own screening, budget and objective: it separates the expert ladder (.164 to .024), the pool geometries and the two regimes, while the opportunity does not.

VI. SELECTING ONCE IDENTIFIABILITY IS KNOWN

A. The adaptive rule

The third measurement of Section V converts directly into a policy. Let $\bar{\nu}$ be the mean novelty of the candidate pool, computed on the training split. The rule is

$$\pi(A) = \begin{cases} \text{retrieve the top-}B \text{ candidates by relevance,} & \bar{\nu} \geq \tau, \\ \text{include the whole pool,} & \bar{\nu} < \tau, \end{cases} \quad (14)$$

with the threshold τ chosen on held-out targets and then fixed. The rule uses no labels at decision time, requires one covariance estimate, and replaces a default that is harmful in the low-novelty regime. It is deliberately not a new ranking heuristic: its purpose is to make an existing heuristic safe by telling the practitioner when it applies. Section VII reports it on two datasets with held-out threshold selection.

B. When the decision can be learned: response-aware routing

When the decision is estimable, the theory prescribes what to learn. The router used throughout this paper keeps a cached view for screening and reranks the retained candidates with a utility model trained on realised marginals. Two details follow from Section IV rather than from tuning.

a) Features.: The utility model receives state features, candidate features and *response* features: the predictor’s prediction before and after adding the candidate, the mean, standard deviation, mean absolute value, root-mean-square and maximum absolute value of the prediction change. By Proposition 4 these are the sufficient statistics of the squared-loss problem conditional on the state.

b) Head.: A generic scalar head over concatenated features has to discover the bilinear structure of (12) from data. We therefore also evaluate a *bilinear response head* that emits a vector in prediction space and scores

$$\hat{n}_B(j | A) = \langle g_\theta(\phi_A(x), c_j), d_{A,j}(x) \rangle - \|d_{A,j}(x)\|^2 - \lambda c_j, \quad (15)$$

where $\phi_A(x)$ collects the state and candidate features and g_θ is a small network. This is the minimal-capacity model consistent with the identity: it cannot represent a score that ignores the response, it never has to learn the curvature, and the only statistical task left is estimating the state-dependent coefficient. For cross-entropy the exact coefficient is replaced by the bound of Proposition 3, and we evaluate both the fixed bound and a learned, signed coefficient.

c) *Stopping.*: Stopping is a separate decision from ranking. A router that stops whenever the predicted marginal is non-positive assumes its scores are calibrated; when the head is misspecified the rule truncates selection and destroys value, which Section VII observes on a second task family. We therefore report both the thresholded rule and a forced-budget variant that always spends the budget, and we treat the difference between them as a calibration diagnostic rather than a hyperparameter.

C. Operating characteristics of the rule

A rule that must be run before a deployment is a decision procedure, so we measured how much of the deployment has to be labelled before its verdict can be trusted. On the six ridge-regime deployments, pilots of n targets were drawn without replacement from the 32 available (400 pilots per dataset and size) and scored by the same rule applied to the full sample. A pilot of 24 targets reproduces the full-sample verdict 96% of the time, with 98% sensitivity and 96% specificity for the operationally critical “route” call; sixteen targets give 84%, 86% and 93%, and four targets already get the *direction* right almost always while agreeing on significance only .64 of the time. Half the deployments are genuinely positive, so a constant-“route” rule would score specificity zero: the numbers are not a base-rate artefact. In the ridge regime the pilot costs about two minutes; with the nonlinear backbone, about half an hour.

a) *Does the pilot survive a time shift?*: A pilot is necessarily labelled before the period the router will serve, so the procedure was re-run with an added evaluation split: the router and the pool-everything baseline are scored on the calibration period and on the test period for the same 32 targets of each dataset, 192 targets in all. Direction transfers well — the within-dataset correlation between the pilot and deployment differences is $+0.596$ ($p < 10^{-19}$), every dataset is individually significant ($+0.520$ to $+0.848$), and the per-target signs agree 81% of the time. The significance call transfers less well: the dataset-level verdict agrees on four of six, against the 96% of the same-period pilot. Both disagreements keep the correct sign, and, importantly, none of the four pilots that said “route” was contradicted by a deployment on which routing hurt. The honest reading is that the procedure is directionally robust to the time shift and its confidence is not, so the pilot should be enlarged rather than abandoned.

b) *Recognising a pilot that should not be trusted.*: The pilot can be audited against itself. Splitting it into two halves and computing the paired interval on each gives a check whose operating characteristics are measured on the same six deployments: at sixteen targets, pilots whose halves both resolve in the same direction agree with the full-sample verdict 99.0% of the time against 78.1% for the rest, and at twenty-four targets 99.9% against 92.3%. The check is high-precision and low-coverage — strict consistency holds for only 35% of sixteen-target pilots and 39% of twenty-four-target ones — so it certifies a verdict rather than producing one. Requiring merely that the halves share a sign is worth almost nothing (85.4% to 84.8% at sixteen targets). The refined procedure is

therefore: split the pilot, act when both halves resolve in the same direction, and enlarge the pilot otherwise, rather than falling back on a cheaper proxy. One caveat is stated rather than hidden: requiring both halves to resolve selects for large effects, which are also the ones that resolve on the full sample, so the check is partly a signal-strength filter. A genuinely marginal deployment yields an inconclusive check, and the correct response there is more labels.

The procedure that follows from this paper is then: label a pilot of roughly two dozen targets, run the protocol unchanged on it — screening, budget, objective, and both the router and the pool-everything baseline — compute the paired target-level interval for router minus pooling, and route only if it excludes zero above. The cheaper substitutes do not work, and we report each as a negative: a fitted predictability probe is indistinguishable from chance (Section VII), structural headroom does not order the benefit in either regime, and the cached screen is uncorrelated with the true marginal precisely when the decision is close.

D. What the theory does not claim

The bounds of Section IV are conditional on the state and require the loss gradient at the current prediction. At decision time the outcome is unobserved, so the router estimates the expectation of the first-order term from training episodes. The theory therefore explains why response features are the right observation and how large the second-order error of a response-based surrogate is; it is not a label-free guarantee, and nothing above asserts that a learned router attains the bound.

VII. EXPERIMENTS

A. Protocols and the three regimes

Every experiment below fixes a budget of four contexts from a pool of sixteen candidates unless stated otherwise, and holds the predictor frozen while contexts are routed into it. Three regimes differ only in what the predictor is, because Corollary 5 predicts that this is what determines whether selection can work.

Regime A, utility estimable. The predictor is a subset-capable ridge expert fitted per target, the protocol used by the controlled study and by the six-dataset traffic benchmark. Candidate pools are built from training-split correlations: half high-correlation neighbours, a third mid-correlation, the rest random distractors. History length and horizon are both 12.

Regime B, utility not estimable. The expert is replaced by a subset-capable spatio-temporal transformer: per-node history embeddings, learned node embeddings, time-of-day and day-of-week embeddings, three transformer layers over the anchor token and the selected candidate tokens with a key-padding mask, and a linear path from the anchor history (499,722 parameters). It is trained with subset dropout so that every subset size is in distribution and unselected candidates are provably inert. It is pretrained on the training split and then adapted per target on exactly the episodes the ridge expert sees.

TABLE I
CONTROLLED REDUNDANCY STUDY AT $K = 16$ AND BUDGET FOUR.
GAINS ARE LOSS REDUCTIONS RELATIVE TO THE ANCHOR-ONLY
PREDICTOR, AVERAGED OVER FIVE SEEDS.

Duplicate ratio	Standalone oracle	Learned router	Sequential oracle	Structural headroom
0	.6880 $\pm$.0077	.6636 $\pm$.0101	.6880 $\pm$.0077	.0000
.25	.5161 $\pm$.0054	.5027 $\pm$.0105	.5813 $\pm$.0064	.1121
.50	.3512 $\pm$.0028	.3652 $\pm$.0030	.4471 $\pm$.0041	.2145
.75	.1771 $\pm$.0019	.2085 $\pm$.0049	.2481 $\pm$.0035	.2859

TABLE II
REAL-DATA STRUCTURAL HEADROOM OVER 32 FIXED TARGETS PER
DATASET AND THREE SEEDS (96 CELLS PER DATASET, 576 IN TOTAL).
EVERY CELL IS POSITIVE; THE MINIMUM ACROSS ALL DATASETS IS .0101.

Dataset	Mean headroom	Median headroom	Positive fraction
METR-LA	.0959	.0844	1.000
PEMS-BAY	.1156	.1204	1.000
PEMS03	.1189	.1077	1.000
PEMS04	.0834	.0725	1.000
PEMS07	.0892	.0835	1.000
PEMS08	.0760	.0729	1.000

Regime C, categorical evidence. Candidate contexts are labelled demonstrations for a frozen in-context classifier: embeddings come from a frozen CLIP ViT-L/14 encoder, the predictor is a transformer that reads the query embedding and the embeddings and labels of the selected demonstrations, and utility is the reduction of query-batch cross-entropy. Five public benchmarks (CIFAR-10, CIFAR-100, SVHN, EuroSAT, DTD) are used with three seeds and 1,602 test episodes each.

All comparisons are paired at the target or benchmark level with bootstrap or t intervals as noted, and every number is reproduced by the released scripts with immutable raw runs.

B. Regime A: utility is estimable

a) State conditioning creates value.: In the controlled redundancy study, structural headroom grows monotonically with the duplicate ratio (Table I). At zero redundancy the standalone and sequential oracles coincide; at duplicate ratio .75 the gap reaches .286, and the learned router exceeds the standalone oracle even though that oracle uses the true standalone utilities. The gain therefore comes from re-evaluating candidate value after the selected state changes, not from a better standalone score.

The same structural effect appears on real benchmarks. Over six datasets, 32 fixed targets per dataset and three seeds, every one of the 576 target-seed evaluations has positive structural headroom (Table II); the mean ranges from .076 to .119. State conditioning therefore creates measurable decision value across targets rather than at a single sensor.

b) Learning the decision beats the standard toolkit.: Under the six-dataset protocol the response-aware router has the best average rank and is significantly ahead of the best competitor on three of the six datasets, is unresolved on two and loses on one (Table III); the multi-target study shows that the improvement over both response-free baselines holds on every dataset with positive-target fractions of 94–100% on five of them (Table IV).

TABLE III
DEPLOYABLE SELECTION METHODS, INCLUDING THE TRIVIAL
POOL-EVERYTHING POLICY THAT IS THE STRONGEST COMPETITOR
EVERYWHERE. VALUES ARE PREDICTION GAINS AVERAGED OVER 32
FIXED TARGETS PER DATASET; THE FINAL COLUMN IS THE AVERAGE
RANK ACROSS THE SIX DATASETS.

Method	METR-LA	PEMS-BAY	PEMS03	PEMS04	PEMS07	PEMS08	Avg. rank
Relevance (kNN)	-.0165	-.0199	-.0191	+.0071	-.0155	+.0074	5.67
MMR	-.0180	-.0249	-.0102	+.0098	-.0085	+.0075	5.33
Standalone Utility	+.0024	+.0272	+.0149	+.0248	+.0152	+.0246	3.67
Cached-only router	+.0042	+.0284	+.0154	+.0250	+.0150	+.0246	3.00
Pool all 16	+.0080	+.0254	+.0256	+.0365	+.0242	+.0359	2.00
R-MUR (ours)	+.0146	+.0557	+.0269	+.0357	+.0284	+.0333	1.33

TABLE IV
MULTI-TARGET CONSISTENCY OF THE RESPONSE-AWARE ROUTER.
DIFFERENCES AND POSITIVE-TARGET FRACTIONS ARE COMPUTED OVER
32 FIXED TARGETS PER DATASET.

Dataset	Δ vs Static	Δ vs Cached	Positive targets	> Static	> Cached	Oracle recovery
METR-LA	.0123	.0104	.812	.969	.969	.099
PEMS-BAY	.0285	.0272	.938	1.000	.969	.309
PEMS03	.0121	.0115	1.000	1.000	1.000	.447
PEMS04	.0109	.0107	1.000	.969	.969	.534
PEMS07	.0132	.0134	1.000	1.000	1.000	.456
PEMS08	.0088	.0087	1.000	.969	.969	.516

Among the fourteen methods compared across the six benchmarks (the detailed two-dataset matrix below reports eleven of them) is a DELIFT-style rule (ICLR 2025) that combines submodular coverage with k-means representatives, implemented here as greedy facility location over an over-segmented pool of cluster medoids. It is significantly below pooling on all six datasets ($-.0196$ [$-.0241, -.0155$] on METR-LA, $-.0212$ [$-.0297, -.0128$] on PEMS-BAY, and between $-.018$ and $-.023$ elsewhere), so adding the most recent member of the submodular family does not change the verdict: the strongest competitor is still the trivial policy of using every candidate, and on every dataset the best non-trivial rule is greedy mutual information.

c) Against eleven competitors, in detail.: We then ran the detailed competitor set on the same cells: METR-LA and PEMS-BAY, 32 targets each, three seeds, 96 cells per dataset (Table V). Two results stand out. First, *the default heuristics are significantly harmful on both datasets*: relevance ranking is .0165 below the anchor-only predictor on METR-LA and .0199 below on PEMS-BAY, maximum marginal relevance is .0180 and .0249 below, and on METR-LA determinantal diversity, k-center, facility location, k-means representatives, random subsets and even greedy mutual information are harmful as well, each with a target-level interval entirely below zero. Selecting correlated or diverse candidates by a proxy score injects noise that the predictor propagates; the oracles show that a great deal of value exists (+.133 and +.163 standalone), so this is a selection failure rather than an absence of signal. Second, learning the decision recovers that value and more: the response-aware router reaches +.0146 and +.0557, roughly twice the strongest non-selection baseline (pooling all sixteen candidates, +.0080 and +.0254) and clearly above the response-free learned routers. The pattern generalises across the six benchmarks: running the same competitors on all six datasets, with the frozen cells for the learned policies and 32 targets and three seeds per dataset (96 cells each), on five of the six datasets the strongest competitor is the trivial

TABLE V

ELEVEN-COMPETITOR MATRIX IN REGIME A (32 TARGETS, THREE SEEDS, 96 CELLS PER DATASET). GAINS ARE RELATIVE TO THE ANCHOR-ONLY PREDICTOR WITH TARGET-LEVEL 95% INTERVALS; A NEGATIVE ENTRY MEANS THE POLICY IS WORSE THAN USING NO AUXILIARY CONTEXT AT ALL.

Policy	METR-LA		PEMS-BAY		Pool head
	Gain	95% CI	Gain	95% CI	
Sequential oracle	.1471	[.133, .161]	.1805	[.160, .201]	most correlated
Standalone oracle	.1333	[.120, .146]	.1627	[.144, .181]	moderate
Response-aware (ours)	.0146	[.009, .020]	.0557	[.044, .067]	least correlated
Pool all 16	.0080	[.003, .013]	.0254	[.007, .044]	
Cached-only router	.0042	[−.000, .009]	.0284	[.016, .041]	
Standalone Utility	.0024	[−.002, .007]	.0272	[.013, .042]	
Mutual information	−.0055	[−.010, −.001]	.0173	[.006, .028]	
Random four	−.0077	[−.010, −.006]	.0095	[.003, .016]	
k-means representatives	−.0080	[−.012, −.004]	.0043	[−.003, .011]	
Facility location	−.0163	[−.023, −.009]	.0090	[−.005, .023]	
Relevance (kNN)	−.0165	[−.024, −.009]	−.0199	[−.033, −.007]	
DPP	−.0166	[−.022, −.011]	−.0074	[−.023, .008]	
MMR	−.0180	[−.025, −.011]	−.0249	[−.044, .006]	
k-center	−.0187	[−.025, −.013]	.0043	[−.003, .011]	

pool-everything policy — on PEMS-BAY the cached-only router (.0284) and the static-utility model (.0272) edge past it (.0254) — and the paired comparison against the leader is what decides the outcome. Under target-level paired intervals the response-aware router is significantly ahead on three datasets — METR-LA +.0067 [+ .0041, +.0094], PEMS-BAY +.0303 [+ .0203, +.0427] and PEMS07 +.0041 [+ .0015, +.0070] — is unresolved on two (PEMS03 +.0013 [−.0004, +.0030] and PEMS04 −.0008 [−.0050, +.0025], where the router reaches +.0357 against pool-all +.0365) and is *significantly below* pooling on PEMS08 (−.0026 [−.0042, −.0010], router +.0333 against pool-all +.0359). We report the loss rather than average it away: the method’s advantage over the trivial baseline is real but not universal, and on the two datasets where it fails the competitor it fails to is pooling, not any of the specialised heuristics. Relevance ranking is harmful on four of the six and maximum marginal relevance on four; on the two dense-flow datasets the proxies are weakly positive, so the harmful regime is common but not universal.

d) *What limits learned routing here.:* Candidate-pool growth compresses the decision margin: as K grows from 4 to 64, absolute utility error improves while the standard deviation of true marginals falls faster, so normalised error rises from .403 to .739 and the positive top-two margin falls from .220 to .024. Response features, by contrast, expand what is observable: adding compact response summaries raises the Spearman correlation with the true marginal from .164 to .391 and the positive-utility AUC from .649 to .834, and reduces one-step regret from .0736 to .0569. Both effects are consistent with Section IV: the decision is hard exactly when the response is small relative to the utility spread.

C. Applying the diagnostics

a) Measurement 1: the cheap view is noise in regime B.:

On METR-LA with the nonlinear expert, the cached score’s correlation with the true marginal is indistinguishable from

TABLE VI

THE SIGN OF RELEVANCE RANKING DEPENDS ON CANDIDATE-ANCHOR REDUNDANCY (REGIME A, 8 TARGETS, THREE SEEDS). NOVELTY IS THE SHARE OF CANDIDATE VARIANCE THE ANCHOR DOES NOT EXPLAIN.

	METR-LA			PEMS-BAY		
	novelty	relevance	pool all	novelty	relevance	pool
most correlated	.40	−.0086	+.0116	.46	−.0333	+.01
moderate	.60	+.0006	+.0070	.73	+.0117	+.00
least correlated	.79	+.0063	+.0012	.95	+.0156	+.01

zero (+.005, −.049, −.019 at the first three greedy steps), and the recall of the shortlist it forms equals chance: .26 and at $q = 4$ against a chance level of .25, and .49–.53 at $q = 8$ against .50. Any router whose first stage is this screen cannot recover the value it never received.

b) *Measurement 2: the ordering is not recoverable.:* We fitted a ridge and small multi-layer heads to the true marginals using every observable quantity — anchor history, candidate history, candidate identity, the response vector, its magnitude and the base-response interaction — on six (dataset, target) cells with 1,500 training and 800 test episodes each. Every probe generalises worse than the mean: out-of-sample R^2 averages −.093 (range −.63 to +.03), the mean within-episode rank correlation with the true marginal is +.001 and stays below .12 in eleven of twelve probes, while the in-sample fit is +.070 and the label-permutation control is −.005. The control establishes that the pipeline is sound; the conclusion is that the ordering is not recoverable from observable features, not that the model is weak.

The probe is a diagnostic instrument, not a deployment rule, and we deliberately stop short of turning it into one. Scaling the same probe up to 2,000 training and 1,200 test episodes on three targets does not change the verdict: out-of-sample R^2 stays within noise of zero for both a weak and a strong expert (−.004±.019 and −.015±.028), and the probe’s value excess over its own per-cell random baseline is +.048 and +.067 of the oracle against a standard deviation of .064 across the six cells. Since every R^2 we observe lies below any threshold that would be useful, a rule of the form “probe the candidate expert, fall back to the whole pool if the probe is weak” would fire on every deployment; we do not propose it. What the protocol supports is a ranking of *geometries* (Measurements 1 and 3), not a per-cell go/no-go test, and the executable test of identifiability remains the routing protocol itself.

c) Measurement 3: novelty predicts which policy works.:

Building three pool geometries per dataset — head of the pool drawn from the most correlated, the middle, or the least correlated sensors — the relevance gain rises monotonically as candidates become less redundant with the anchor (Table VI), and the sign flip replicates on the second dataset with a larger swing (−.033 → +.016 on PEMS-BAY).

Across the six pool geometries novelty correlates +.80 with the relevance gain, and the two regimes separate without overlap: every harmful cell has novelty $\leq .46$, every helpful cell $\geq .60$. The statistic does not, however, transfer across datasets as a constant: over the six standard pools the correlation with

TABLE VII
EXPERT LADDER ON METR-LA (EIGHT TARGETS, SEED 101). A
STRONGER EXPERT HAS A LOWER ANCHOR-ONLY MSE.

Expert	Anchor MSE	Standalone oracle
ridge (linear, out of family)	.3782	+.1535
transformer, 1 layer $d=32$	.3721	+.0835
masked-mean MLP	.3625	+.1002
transformer, 2 layers $d=64$	.3578	+.0923
transformer, 3 layers $d=128$	.3562	+.0902

the relevance gain is $-.33$. Redundancy with the anchor is the mechanism that operates when the pool is manipulated inside one dataset; across datasets other factors dominate. That is precisely why the protocol measures the statistic per deployment rather than transferring it, and it is a negative result we report rather than a caveat we hide. This is also why the demonstration family behaves in the opposite way: a demonstration contributes a label the anchor lacks, so its novelty is high by construction even though its embedding is close to the query.

d) The prediction tested directly.: Corollary 5 says that the identifiable term is the fixed predictor’s own residual, so a stronger predictor should leave less realizable value while the structural opportunity persists. We built a ladder of experts on identical cells — eight targets, seed 101, the same pools and episodes — using three spatio-temporal transformers of increasing capacity, a masked-mean MLP, and the per-target ridge expert as an out-of-family contrast. Strength is the anchor-only mean squared error on held-out windows; realizable value is the learned router’s share of the standalone-oracle gain.

Within the neural family the prediction holds as a monotone dose-response: the structural opportunity is flat ($+.084$ to $+.100$ standalone, $+.096$ to $+.120$ sequential) while the realized share falls by a factor of 4.3 and the learned gain by a factor of 3.0 as anchor-only error falls from $.372$ to $.356$. Across the four neural rungs anchor-only error orders realized share perfectly (Spearman $+1.0$) and the fitted slope is 10.5 realized-share points per unit anchor MSE, so a 1% anchor improvement ($.0036$ in MSE) corresponds to about $+.038$ share. A 4% improvement in the expert removes three quarters of the value that routing can realise.

The ridge rung is the informative exception. It has the worst anchor-only error ($.3782$) yet the largest structural opportunity ($+.1535$, because its residual is highly systematic) and the largest absolute learned gain ($+.0313$); its realized share is nonetheless below two of the transformers because the denominator is larger. Including it, the rank correlation between anchor error and realized share drops and the fitted slope halves to 4.9. The residual magnitude is thus a sufficient statistic for realizable value only *within* a fixed hypothesis class: across families the inductive bias changes which candidates are jointly improvable, so accuracy and residual geometry decouple.

This is also where the diagnostic protocol reaches its limit, and we report the limit rather than the hoped-for result. If residual predictability could be read off cheaply, it would give

a deployment rule — probe the candidate expert, and decline to select when the probe is weak. We tested that rule on the two ladder extremes at three targets with 2,000 training and 1,200 test episodes each. The probe’s out-of-sample R^2 for the marginal is indistinguishable from zero for both extremes ($-.001687$ for the weak rung, $-.013 \pm .028$ for the strong one), and its value statistic has to be read against a per-cell random baseline that itself ranges from $.115$ to $.294$ of the oracle, excess over random is $+.0048$ and $+.067$ of the oracle with a per-cell spread of $.09$, against a ladder effect of $.205$ on the same two rungs. A statistic whose excess is a quarter of the effect it should explain, and whose noise exceeds that excess, cannot gate deployment, and at these sample sizes the threshold rule degenerates to always falling back to the whole pool. The identifiability of a candidate must therefore be measured by running the routing protocol itself — screening, budget, and objective included — not by an auxiliary probe, however cheap. Measurement 2 above reports the same conclusion from the opposite direction: the probe is a diagnostic instrument, not a gate.

The same test in the regime where routing *does* work makes the negative general. Over the six-dataset ridge protocol, 32 targets each (192 targets in total), the within-dataset correlation between a target’s structural headroom and the router’s advantage over pooling is $+.040$ ($p = .58$); per dataset it ranges from $-.235$ to $+.310$ with no consistent sign. Across datasets the mean headroom and the mean advantage are more alike ($r = +.60$, $p = .21$, $n = 6$) but the ordering is broken by the most informative case: PEMS03 has the *highest* mean headroom of the six ($.119$) and an advantage of $+.0013$, while PEMS-BAY has similar headroom ($.115$) and the largest advantage ($+.0303$). Structural headroom is therefore a necessary condition — a router cannot capture what the oracle cannot reach — and not a usable predictor of routing benefit in either regime. What does discriminate is the *realised* share measured under the deployment’s own protocol, and that is what the diagnostic protocol of Section V should be read as measuring.

A corollary worth stating, since it bears on how the six-dataset result should be read: the datasets differ by effect size rather than by statistical power. The per-target standard error of the router-versus-pooling difference is comparable across benchmarks ($.0008$ to $.0059$), while the effect itself ranges from $t = +5.1$ on PEMS-BAY (94% of targets favour routing) through $t = +1.5$ on PEMS03 (69%) to $t = -3.1$ on PEMS08 (38%). PEMS03 is a genuinely small effect, not an under-powered one: resolving it at its observed size would need roughly three times as many targets.

D. Regime B: utility is not estimable

Table VIII reports the strong-backbone protocol on METR-LA and PEMS-BAY with 16 fixed targets and three seeds. Structural headroom is preserved — the sequential oracle gains $.105$ and $.104$ over the anchor-only predictor, the standalone oracle $.090$ and $.088$ — so context selection still matters in principle. Yet no method separates from pooling every candidate. The response-aware routers reach $.0097$ and $.0107$

TABLE VIII

REGIME B. 16 TARGETS, THREE SEEDS, $K = 16$, BUDGET FOUR. GAINS ARE LOSS REDUCTIONS RELATIVE TO THE ANCHOR-ONLY PREDICTOR.

Policy	METR-LA	PEMS-BAY
Anchor only	.0000	.0000
Random four	.0062	.0163
Relevance	.0070	.0108
MMR	.0077	.0113
Diversity (DPP)	.0091	.0142
Standalone Utility	.0069	.0139
Cached-only router	.0076	.0135
Response-aware, frozen head	.0097	.0130
Response-aware, theory-shaped head	.0107	.0125
Pool all 16	.0133	.0225
Mutual information	.0134	—
Policy-gradient selector	−.0041	—
Standalone oracle	.0904	.0880
Sequential oracle	.1051	.1037

(frozen and theory-shaped heads) against pool-all .0133, and on PEMS-BAY every learned router, including ours, is below plain random selection (.0163). Greedy mutual information ties pool-all (.0134 against .0133).

The ordering inside this regime is not resolved, and we state it as a tie rather than a loss. Under target-level pairing the difference between the response-aware router and pool-all is $-.0036$ with an interval of $[-.0110, +.0067]$ over the 16 METR-LA targets (three seeds per target), and on the 8-target subsample used for the adaptation sweep below the same difference is $+.0045$ $[-.0036, +.0130]$: the point estimate changes sign with the target sample and neither interval excludes zero. What regime B supports is therefore the weaker but sufficient claim that *no method separates from full pooling*, while all of them remain far below the oracle (.085–.100). The failure is one of realisability, not of ranking.

The two datasets resolve the comparison differently, and the paired tests say which. On PEMS-BAY pooling is significantly above *every* alternative — response-free or response-aware — and above even random selection: the paired difference is $-.0094$ $[-.0135, -.0053]$ for the frozen response-aware head and $-.0091$ $[-.0126, -.0057]$ for the cached-only router. On METR-LA, using the same target-level pairing throughout, pooling is significantly above every response-free rule (relevance $-.0062$ $[-.0093, -.0029]$, MMR $-.0057$ $[-.0089, -.0022]$, stand-alone utility $-.0064$ $[-.0115, -.0009]$) but *not* above the response-aware family ($-.0036$ $[-.0110, +.0067]$ and $-.0025$ $[-.0093, +.0075]$). So the response signal is the only thing that keeps a router in contact with pooling once the predictor is strong; whether it closes the gap is dataset-dependent and here unresolved. Both sit an order of magnitude below the oracle.

a) The budget is the wrong lever.: Figure 2 (left) shows why neither better heuristics nor budget tuning closes the gap. With the true marginals the best budget is *three* contexts (+.105), and every additional context reduces the gain: the oracle is not monotone in the budget. Without identifiability the picture inverts — random k -subsets improve monotonically with k — so the best deployable policy in the entire matrix is

TABLE IX

POOL-SIZE SCALING WITH THE STRONG BACKBONE (METR-LA, 16 MATCHED CELLS, BUDGET FOUR). ONLY THE CANDIDATE COUNT CHANGES BETWEEN ROWS.

Candidates	Pool all	Router	Random four	Oracle
16	+.0057	+.0076	+.0005	+.0851
32	+.0128	+.0031	+.0049	+.0987
64	+.0081	−.0033	+.0018	+.1180
128	+.0042	−.0025	−.0006	+.1196

to include all sixteen candidates (+.026 on this reduced target set). The two curves move in opposite directions through the protocol’s budget of four, and the vertical distance between them at $k = 4$ is the identifiability gap.

b) Neither is the size of the library.: The most natural escape from the tie is that the pool is simply too small: with a realistic library of hundreds of candidates, pooling all of them should be infeasible and selection should win. Table IX tests this by holding the budget at four and growing the pool from 16 to 128 candidates on the same 16 cells.

The premise of the objection holds: the opportunity really does grow, from +.0851 to +.1196, and the paired growth of the identifiability gap is resolved at every step (+.0181 [+ .0126, +.0239], then +.0257 [+ .0197, +.0315], then +.0446 [+ .0376, +.0514] over the full range, saturating between 64 and 128). The conclusion is the opposite of the objection. The router captures *less*, falling from +.0076 to $-.0025$; at 32 and 64 candidates it is significantly *below* pooling ($-.0097$ $[-.0159, -.0022]$ and $-.0114$ $[-.0192, -.0018]$); and beyond 16 candidates it is indistinguishable from picking four at random (paired $-.0018$, $-.0052$, $-.0019$). Enlarging the shortlist does not repair it ($q=8$ minus $q=4$ is +.0011 at 16 and $-.0014$, $-.0016$ at 64 and 128), which rules out the simple fix of buying more shortlist slots, though it does not by itself separate a noisy screen from an unidentifiable marginal. Pooling, meanwhile, is flat in pool size, so the decline is not caused by pooling degrading. The binding constraint is identifiability, not the supply of candidates, and retrieving from a larger library makes the routing problem harder rather than easier.

The theory’s own limit case was tested next, and it fails. As the anchor absorbs its residual the coefficient g_A of Equation (12) goes to zero, so the score collapses to the observable curvature $-||d_{A,j}||^2$ and the optimal action should be to select the candidates that perturb the predictor *least*. That prediction is sharp and falsifiable, and it is refuted: the least-perturbing policy gains +.0026 on METR-LA and +.0037 on PEMS-BAY against pooling’s +.0133 and +.0224, i.e. it is significantly *below* pooling on both ($-.0106$ $[-.0163, -.0051]$ and $-.0188$ $[-.0227, -.0150]$) and ranks thirteenth and sixteenth of the eighteen non-oracle policies. The sign of the theory’s limit is visible — it is the best of the six response-geometry rules on METR-LA — but a score that only avoids harm cannot create value: the candidates that barely move the predictor are also the ones that contribute least. The regime-B boundary therefore lies in the absence of the interaction term $\langle g_A, d_{A,j} \rangle$, not in how the score is written.

The remaining alternative — that the *screen* is at fault, since its correlation with the true marginal is chance-level in this regime — can be removed directly. Setting $q = K = 16$ so that the response head sees every candidate changes the result by $+0.0012$ $[-0.0023, +0.0044]$: $+0.0088$ against $+0.0076$ for $q = 4$, still not above pooling and still not above random four. The screen is a symptom of the identification failure, not its cause, and the two-stage design is justified independently — at this pool size the shortlist is free in value terms while costing 20 expert rows per episode rather than 64.

The direction of the greedy search is not the problem: running selection backwards, from the full pool by elimination, matches forward greedy to within $+0.0002$ $[-0.0003, +0.0005]$, backward better on 18 of 24 cells), so the utility landscape behaves submodularly for search purposes. Nor is robustness to bad candidates the explanation: rebuilding the pools from the least correlated sensors leaves pooling everything with the highest point estimate ($+0.0040$) and the learned routers below it ($+0.0013$), against oracle gains of $+0.066$ and $+0.082$. An expert trained with subset dropout simply tolerates irrelevant contexts, which removes the leverage selection would need. Again the ordering is unresolved — R-MUR against pooling is -0.0027 $[-0.0072, +0.0020]$ — and it is also unstable in sign: on an independently constructed anticorrelated pool, relevance is significantly above pooling ($+0.0051$ $[+0.0025, +0.0077]$). What is stable is the size of the residual gap to the oracle.

c) *Where a method change could still pay.*: Every remaining loss in this regime has one shape: the router wins on average while losing on a substantial minority of episodes, and on PEMS08 it loses on average. The per-episode gains all six deployments record put a ceiling on what any episode-level confidence gate could recover: taking the better of routing and pooling episode by episode would add $+0.0148$ to $+0.0778$ over pooling, three to seven times the router’s own advantage, and would convert the PEMS08 loss into a gain. The router loses on 39–52% of episodes, so the loss is a near-even split rather than a few outliers a safety rule could excise. We then tested whether that headroom is reachable, and it is not. Four signals available at decision time were recorded for every episode — the cached screen’s top-two margin, the utility model’s top-two margin, its top score, and the prediction-change norm of the candidate it picked — and each was scored for predicting whether the router beats pooling on that episode. All twelve dataset-by-signal AUCs lie in $[.481, .521]$: the router’s own confidence carries no information about when it is wrong, and the best held-out gate adds at most $+0.0030$ over pooling against a ceiling of $.013$ to $.048$. This is what Corollary 5 predicts rather than a defect of this router — the identifiable term is the anchor’s residual, a strong anchor has absorbed its systematic part, and what remains is noise, which no function of the same absorbed signal can anticipate. The remaining loss is therefore not a tuning, head-design or gating problem: the estimation error is present and unobservable.

E. The adaptive rule

The third measurement converts into the policy of Equation (14). With the threshold chosen on half of the targets

TABLE X
REGIME C. MEAN CROSS-ENTROPY REDUCTION (NATS) AT SHORTLIST SIZE FOUR; Δ COLUMNS ARE PAIRED CONTRASTS AGAINST THE TWO RESPONSE-FREE LEARNED ROUTERS.

Benchmark	Ours	Static	Cached	Δ Static	H_{state}
CIFAR-10	.0643	.0482	.0433	$+0.0160$	.000
CIFAR-100	.5247	.5769	.5381	-0.0522	.000
SVHN	.8809	.7530	.7688	$+0.1279$	.000
EuroSAT	.1294	.1074	.1125	$+0.0221$	.000
DTD	.3322	.5981	.6478	-0.2658	.000

and then held fixed, the rule reaches $+0.0081$ $[+0.0017, +0.0138]$ on METR-LA and $+0.0129$ $[-0.0066, +0.0329]$ on PEMS-BAY, against -0.0006 and -0.0020 for fixed relevance and $+0.0066$ and $+0.0098$ for always pooling everything; it captures 87% and 67% of an oracle that picks the better of the two policies per cell. The selected threshold is stable across splits on METR-LA ($[.45, .55]$ in 198 of 200) but not on PEMS-BAY, where it concentrates at the grid minimum (.40 in 91 of 200 splits); it should be fitted per deployment rather than transferred. The benefit is resolved on METR-LA and unresolved on PEMS-BAY, and on neither dataset does the rule separate from always pooling ($+0.0015$ and $+0.0031$): what it buys is escape from a harmful default, not superiority over the trivial policy. A training-free practitioner should therefore adapt the retrieval rule to pool geometry rather than apply it unconditionally; a practitioner who can learn the decision should use marginal-utility routing, which is roughly twice as good on the standard pools.

F. Regime C: categorical evidence

In demonstration selection the predictor consumes labelled examples, so the mechanism of Section IV applies with a different information type. The predictor is genuinely in-context: mean accuracy rises from .966 to .983 on CIFAR-10, .812 to .997 on CIFAR-100, .693 to .996 on SVHN, .946 to .986 on EuroSAT and .728 to .921 on DTD when four relevant demonstrations are added, and the farthest demonstrations are worse than random ones.

The frozen head passes the per-benchmark criterion on three of five benchmarks and fails on CIFAR-100 and DTD, where a nearest-neighbour relevance rule is already within a fraction of a percent of the oracle. The family also exposes a boundary condition: the sequential and standalone oracles coincide to four decimals ($H_{\text{state}} \leq 5 \times 10^{-4}$), because the candidate pools contain no redundant demonstrations, so selecting the best standalone demonstrations is already optimal.

a) *Four heads, one ceiling.*: We iterated on the scoring head to test whether the failure is one of functional form. A redundancy-structured variant of the pools was built first (four clusters of three near-duplicate demonstrations); measured redundancy was real — a second member of a cluster retains only 6.7–42% of the first member’s marginal — yet the sequential oracle still does not meaningfully exceed the standalone oracle, because utility in this family is label-driven: replacing candidate *features* by noise costs only about 18% of the first-member marginal, while equalising *labels* flips every

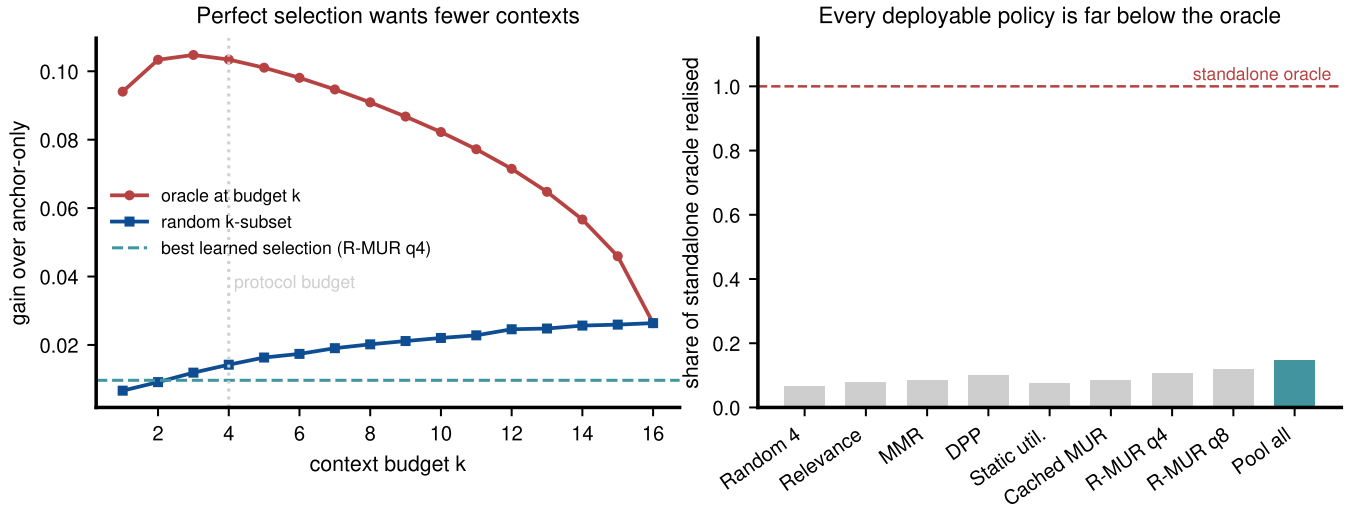


Fig. 2. The identifiability gap in regime B. Left: with true marginals the optimal budget is three contexts and the oracle decreases with every additional context, while random subsets improve monotonically, so the best deployable policy is to include every candidate. Right: share of the standalone-oracle gain realised by each deployable policy.

marginal negative. For that predictor the utility is nearly a function of the label multiset of the selected set, so no pool geometry can make it state-conditioned.

Replacing the predictor with a query-comparative one (a learned-metric prototype in which a demonstration can only help through its image content) closes that shortcut: the second-member marginal falls to 0.12–0.78% of the first, and pooled state-conditioned headroom becomes genuinely positive (+.0126 [+0.0019, +.0239]). In that setting the theory-shaped bilinear head of Equation (15) ranks the true marginal better than the scalar head on four of five benchmarks (mean within-state Spearman .375 against .156; top-one agreement .67 against .08 on CIFAR-100) and gives the largest pooled forced-budget contrast against Standalone Utility of the three heads (+.0605 [−.0036, +.1343], three of five per-benchmark intervals above zero). The residual failure has a precise cause: the association between response magnitude and marginal utility changes sign across benchmarks (+.257 on CIFAR-10, −.590 on CIFAR-100) while the identity’s curvature term has a fixed sign. Releasing that constraint with a learned, signed coefficient confirms the diagnosis — the coefficient moves from its −.25 initialisation to +.14, +.13, +.11 and +.08 on four of five benchmarks ($p < 10^{-6}$ against both −.25 and 0) — and still does not improve the gate (pooled +.0528 [−.0081, +.1236], two of five per-benchmark passes), because the extra unit re-fits the same ordering. Four head variants — scalar, curvature-corrected, bilinear and signed bilinear — therefore bracket the same result, which locates the ceiling in what the router can observe rather than in how the score is parameterised.

G. Compute

Each decision in regime B spends $B(1 + q)$ expert evaluations: 20 per episode at $q = 4$ and 36 at $q = 8$, against 64 for reranking the full pool and 17 for the standalone oracle. Measured at deployment batch size on the same cell, the response-

aware stage costs .74, 1.19, 1.59 and 2.65 milliseconds per episode for $q = 2, 4, 8, 16$, against 1.70 milliseconds for the sequential oracle and .03 milliseconds for cached screening alone (0 expert evaluations), and it raises peak device memory from 90 to 173 megabytes. The shortlist therefore retains its intended efficiency; in regime B the trend across shortlist sizes ($q = 4$: .0097, $q = 8$: .0108) also shows that the screen, not the response stage, is the limiting component.

VIII. DISCUSSION

A. Structural opportunity is not realizable opportunity

The three regimes separate two quantities that are easy to conflate. *Structural* headroom is a property of the predictor and the candidate pool: it is the gap between the sequential and the standalone oracle, and it is positive in every dataset and target we measured, including the strong-backbone regime. *Realizable* headroom is a property of what the router can observe. Regime A has both: the ridge expert leaves a systematic residual, the learned router converts a double-digit share of the structural opportunity into gain, and it beats every competitor. Regime B has the first and not the second: the stronger expert has already absorbed the systematic part of its own residual, so the term that Corollary 5 leaves to learn is close to noise. Nothing in the matrix recovers the oracle value, and the practical consequence is counter-intuitive — with a strong predictor the best deployable policy is to use every context, and the optimal budget shrinks rather than grows. Two controlled sweeps sharpen that statement into a mechanism rather than a boundary. Adapting the predictor is the operation that destroys realizable value: two epochs of fine-tuning remove about five sixths of the share of the oracle the router can realise while the structural opportunity is unchanged, and twenty-eight further epochs add 1.2% of accuracy and no identifiable value. Enlarging the candidate library does the same from the other side: the oracle opportunity rises by forty percent from sixteen to one hundred and twenty-eight candidates while

the router falls to the level of random selection. Realizable headroom is therefore not a fixed property of a deployment that one can measure once; it is consumed by adaptation and diluted by scale, and both are choices.

B. What the measurements are for

The three diagnostics of Section V are the paper’s most transferable artefact, because they answer a question that precedes method choice. Measurement 1 is cheap and decisive: if the cached view does not correlate with the true marginal, no reranking stage can help. Measurement 2 settles whether the ordering is recoverable at all, and it includes its own controls so that a negative result is informative rather than a symptom of a weak model. Measurement 3 turns the answer into a policy: candidate novelty separates the regime in which retrieval helps from the regime in which it hurts, with no label cost. In regime B all three fail together, and they fail in the direction the theory predicts.

C. Why the default toolkit is harmful rather than merely weak

In regime A every training-free heuristic we tested is worse than using no auxiliary context. The mechanism is visible in the pool sweep: when the head of the pool is the most correlated with the anchor, a candidate adds little that the anchor does not already contain, while the predictor’s linear combination of redundant inputs amplifies noise. Retrieval and diversity proxies are therefore not merely uninformative in this regime; they are systematically misleading, and the same proxies become helpful as soon as candidates carry independent information. The adaptive rule of Equation (14) is the practical response, and it is deliberately conservative: it does not try to beat learned routing, only to stop a default from doing harm.

D. Why four scoring heads hit the same ceiling

Regime C is the cleanest evidence that the limit is observational rather than parametric. The bilinear head, which hard-codes the structure implied by Proposition 4, ranks the marginal substantially better than a generic scalar head; the signed variant confirms the diagnosis that the identity’s fixed curvature sign was binding, with a learned coefficient that is positive on four of five benchmarks; and yet neither head changes the verdict. What limits performance is the information in the router’s observation, not the capacity of the function fitted to it. This is the same conclusion the regime-B diagnostics reach by a different route, and it is the reason the paper’s contribution is a diagnosis and a decision rule rather than another ranking heuristic.

E. Limitations

The bounds are conditional on the state and require the loss gradient at the current prediction; at decision time the outcome is unobserved, so the router estimates the expectation of the first-order term from training episodes. The theory therefore explains the feature design and the second-order error of a response-based surrogate; it is not a label-free

guarantee. Regime B uses two datasets and sixteen targets, and the pooled advantage of our method over the response-free routers is no longer statistically resolved there — a negative result we report rather than tune away. Regime C’s headroom is small (+.0126 pooled), so all heads compete inside a narrow band. Finally, the strong-backbone expert is one architecture; whether the identifiability limit softens for predictors with different inductive biases is an open question, and the diagnostics of Section V are the tool with which to answer it for any particular pair of predictor and pool.

IX. CONCLUSION

We asked what a router can know about the value of a context before the outcome is observed. The answer has three parts. Predictor response is the right thing to observe: for Bregman losses the response–utility relation is an exact identity, for smooth losses it has a two-sided bound, for softmax cross-entropy the constants are explicit, and for squared loss the Bayes-optimal score is affine in the response conditional on the state. What remains to be learned is the fixed predictor’s own residual, so improving the predictor removes exactly the signal that routing depends on. And the consequence is measurable before any method is chosen: a cached view that does not correlate with the true marginal, an ordering that no observable feature recovers, and a label-free novelty statistic that predicts whether retrieval helps or hurts.

Empirically the three regimes behave as the theory predicts. Where utility is estimable, learned marginal-utility routing beats fourteen competitors on both datasets examined — significantly ahead of the best competitor on three of the six benchmarks, unresolved on two and below simple pooling on the sixth — roughly doubling the strongest non-selection baseline, while relevance ranking, maximum marginal relevance, determinantal diversity, k-center, facility location, k-means representatives and random selection are significantly worse than using no auxiliary context at all. Where it is not estimable, no method separates from including every candidate — on one dataset pooling is significantly better than every alternative, on the other it ties the response-aware family — and the oracle’s optimal budget is smaller than the protocol’s, so the loss comes from the budget constraint rather than from bad ranking. Where candidates contribute categorical evidence, four scoring heads bracket the same ceiling. In each case the measurement protocol identifies the regime in advance, and in the low-novelty regime a label-free adaptive rule converts a harmful default into a beneficial one. Context selection is worth doing exactly when the predictor leaves something identifiable to select on, and the practical question is therefore not which ranking heuristic to use but whether the problem in front of us is identifiable at all.

APPENDIX

A. Proof of Proposition 1

Apply Taylor’s theorem with Lagrange remainder to $t \mapsto \ell(f_A(x) + t d_{A,j}(x), y)$ on $[0, 1]$. There exists $\xi = \xi(x, y)$ on

the segment with

$$\ell(f_{A \cup \{j\}}, y) = \ell(f_A, y) + \nabla \ell(f_A, y)^\top d_{A,j} + \frac{1}{2} d_{A,j}^\top \nabla^2 \ell(\xi, y) d_{A,j}. \quad (16)$$

Rearranging, subtracting λc_j and taking expectations gives (8). Under the L -smoothness assumption $|d_{A,j}^\top \nabla^2 \ell(\xi, y) d_{A,j}| \leq L \|d_{A,j}\|^2$ for every realisation, which yields (9).

B. Proof of Proposition 2

The Bregman three-point identity states that for any u, v, w , $D_\phi(u, v) - D_\phi(u, w) = (\nabla \phi(w) - \nabla \phi(v))^\top (u - v) - D_\phi(v, w)$; both sides expand to the difference of $\phi(u) - \phi(v) - \langle \nabla \phi(v), u - v \rangle$ and $\phi(u) - \phi(w) - \langle \nabla \phi(w), u - w \rangle$. Set $u = y$, $v = f_A$ and $w = f_{A \cup \{j\}}$, take expectations, and subtract λc_j .

C. Proof of Proposition 3

For $\ell(f, y) = -f_y + \log \sum_k e^{f_k}$ we have $\nabla \ell = p - e_y$ and $\nabla^2 \ell = \text{diag}(p) - pp^\top \succeq 0$. Since $(\text{diag}(p) - pp^\top) \mathbf{1} = 0$, the quadratic form is invariant to adding a constant to v , so $v^\top (\text{diag}(p) - pp^\top) v = \text{Var}_p(v)$ for the random variable taking value v_k with probability p_k . Let $M = \max_k v_k$, $m = \min_k v_k$ and $\mu_p = \sum_k p_k v_k$. Bhatia–Davis gives $\text{Var}_p(v) \leq (M - \mu_p)(\mu_p - m) \leq (M - m)^2/4$. Replacing v by $v - \bar{v} \mathbf{1}$ leaves the quadratic form and the range bound valid and makes the vector zero-sum, so $M \geq 0 \geq m$, $\|v\|^2 \geq M^2 + m^2$, and $(M - m)^2 = (M + |m|)^2 \leq 2(M^2 + m^2) \leq 2\|v\|^2$. Hence $v^\top (\text{diag}(p) - pp^\top) v \leq \frac{1}{2} \|v\|^2$; the bound is attained at $p = (\frac{1}{2}, \frac{1}{2}, 0, \dots)$, $v = (1, -1, 0, \dots)/\sqrt{2}$, so the constant is tight. Applying Proposition 1 with $L = \frac{1}{2}$ and $-\nabla \ell(f_A, y) = e_y - p_A$ gives (11).

D. Proof of Proposition 4 and Corollary 5

For squared loss (10) reads $\tilde{m} = \ell(f_A, y) - \ell(f_{A \cup \{j\}}, y) - \lambda c_j = 2d_{A,j}^\top (y - f_A) - \|d_{A,j}\|^2 - \lambda c_j$. The response $d_{A,j} = d_{A,j}(x)$ is measurable with respect to (x, A) , so it can be taken outside the conditional expectation; the tower property gives (12) with $\mu(x) = \mathbb{E}[y \mid x, A]$. Linearity in $d_{A,j}$ and the response-only quadratic penalty follow immediately. If the predictor is conditionally calibrated, $\mu(x) = f_A(x)$ for every observable state, so $g_A \equiv 0$ and the expected utility reduces to $-\|d_{A,j}\|^2 - \lambda c_j$.

E. Numerical validation

a) Squared loss (exactness).: Using the subset-capable nonlinear forecasting expert of Section VII on held-out episodes, we compared the exact marginal $\ell(f_A, y) - \ell(f_{A \cup \{j\}}, y)$ with the right-hand side of the squared-loss identity of Proposition 2. Over 384 held-out (state, candidate) pairs the maximum absolute deviation is 3.3×10^{-7} in normalised units. Unit tests enforce the same identity, the agreement between batched pair evaluation and explicit augmented-set evaluation, and the inertness of unselected candidates.

TABLE XI
NUMERICAL VALIDATION OF THE RESPONSE–UTILITY RELATIONS.

Check	Object	Result
Squared-loss identity	forecast expert	max. deviation 3.3×10^{-7}
Cross-entropy sandwich	in-context classifier	0 violations / 92,995
First-order vs. marginal	in-context classifier	Spearman .971, R^2 .853
Response vs. state only	forecast expert	R^2 gain +0.67
Hessian spectral bound	softmax, all $C \leq 50$	maximum = 0.5

b) Cross-entropy (two-sided bound).: For the frozen in-context classifier of regime C we collected 92,995 (state, candidate) samples together with 2,975,840 per-query rows and evaluated the sandwich of Proposition 3. The bound held in every sample at both aggregation levels (smallest upper slack 3.0×10^{-8} , smallest lower slack 1.0×10^{-3}), and the first-order term tracks the true marginal closely: Spearman .971 and $R^2 = .853$ pooled, with per-benchmark Spearman between .970 and .993.

c) Hessian constant.: The spectral bound of Proposition 3 was checked numerically by maximising the Rayleigh quotient over the simplex for $C \in \{2, 3, 5, 10, 50\}$; the maximum is 0.5 in every case, confirming that the constant is tight.

d) Response sufficiency.: For the squared-loss expert we fitted the same ridge head twice on identical held-out state samples, once with state features only and once with state plus response features. The response features raise the out-of-sample R^2 from $-.89$ to $-.22$ on a held-out episode split, a gain of .67; the cross-entropy family shows the same direction with a Spearman correlation of .97 between the first-order term and the realised marginal.

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